INCH-POUND

MIL-C-83996A (USAF)
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SUPERSEDING
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20 JUNE 1979

MILITARY SPECIFICATION
CARTRIDGE, DESICCANT

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers one type of desiccant cartridge for use in airborne pneumatic systems.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and Standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS
MILITARY

DOD-D-1000	Drawings, Engineering, and Associated Lists
MIL-D-3716	Desiccants, Activated for Dynamic Dehumidification
MIL-G-5514	Gland Design; Packings, Hydraulic, General Requirements For

STANDARDS
MILITARY

MS-28775	Packing,Preformed, Hydraulic, Plus 275°F, (O-ring)
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-810	Environmental Test Methods

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMIRBC, Robins AFB, GA 31098-5609 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 4440

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The cartridge shall consist of the following major components:

Item No.	Quantity	Item Name
1	1 ea	Container
2	2 ea	Filter
3	See 4.7.5	Desiccant

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials. All materials used in the construction of the cartridge shall be capable of withstanding operating temperatures from -65° F to +160° F without physical or chemical deterioration. The materials shall also be suitable for use in an environment consisting of air, water vapor, and oil vapor or any combination of these.

3.4.1 Container. The container or body of the cartridge shall be constructed in accordance with Figure 1 and shall be protected against deterioration when exposed to climatic and environmental conditions of 3.4 in a manner that will in no way prevent compliance with the requirements of this specification.

3.4.1.1 Ends. The container ends shall be equipped with buttons or end closures that are easily removed by either cutting or breaking to allow passage of air through the cartridge.

3.4.2 Desiccant. The desiccant material shall meet the particle size, particle strength, and corrosiveness requirements as applicable to Type II, Grade H or M desiccant in accordance with MIL-D-3716. The desiccant particle size shall be between 8 and 16 mesh. The desiccant shall be mechanically and chemically nontoxic and shall not deliquesce in the presence of water or water vapor. A gas or liquid shall not be produced or cause to be produced by the desiccant material in reaction to water or water vapor or oil.

3.4.3 Filters. Two filters shall be provided in each end. The filters shall be positively affixed in position to prevent misalignment and possible passage of air and desiccant particles around the filter media.

3.4.4 "O-Ring" seal. The O-Ring seal shall conform to MS-28775. "O"-Ring groove diameter in cartridge body shall be .817 – .822 Inch.

3.5 Design and construction. The cartridge shall be designed and constructed to dehydrate air in a pneumatic system with a maximum operating pressure of 3000 psig and a minimum flow rate of 2.0 SCFM.

3.6 Performance.

3.6.1 Moisture removal. The cartridge shall maintain an outlet moisture content not in excess of that equivalent to a dew point of -65° F at 0 psig in a minimum of 75 standard (70° F and 760 mm Hg) cubic feet of air saturated at 130 +5, -0° F and 300 +0, -15 psig, and passed through the cartridge at a flow rate of not less than 1.2 SCFM plus or minus 1 percent under these conditions.

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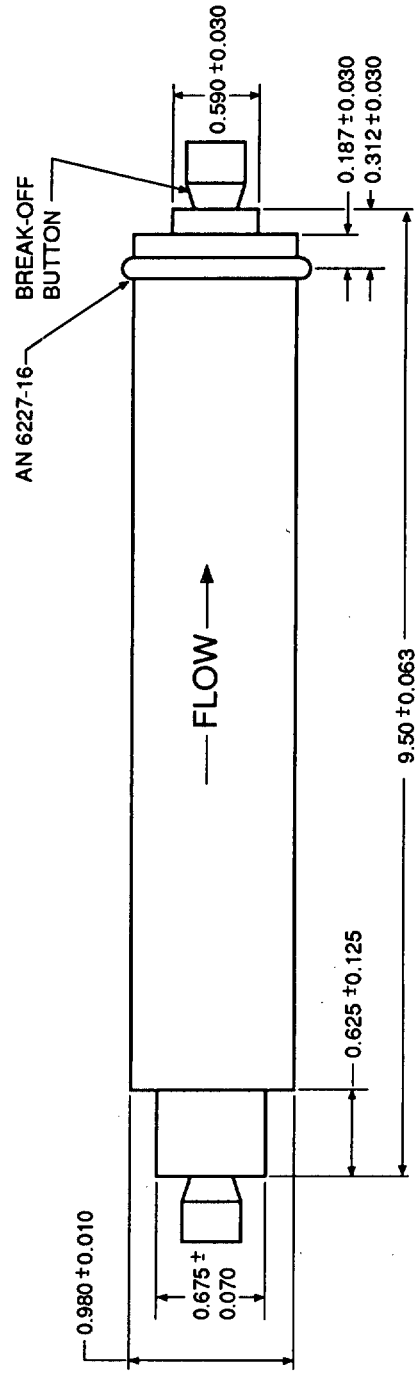


Figure 1.

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3.6.2 Particle filtration. The filter on the output end of the cartridge shall be capable of preventing the passage of desiccant particles 10 microns nominal on the largest dimension. The filter or filter media on the input end of the cartridge shall be capable of preventing the passage of desiccant particles 50 microns nominal on the largest dimension. Or, as an alternate, a 60 mesh screen with filter pad may be used.

3.6.3 Settling and compaction. The cartridge shall be assembled in such manner that the contents of the cartridge shall be settled or compacted so as to prevent free movement of the desiccant.

3.6.4 Leakage. Indication of leakage shall not be evident when a cartridge previously temperature stabilized at 80°F or below is totally immersed for not less than 30 minutes in water maintained at 150° F.

3.7 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DOD-D-1000 shall govern the manufacturer's part number and changes thereto.

3.8 Identification and operational markings. The cartridge shall be marked in accordance with Figure 2. The printing shall be legible and of a permanent nature.

CARTRIDGE, DESICCANT

MFGR_____

MFG PART NO_____

MFG DATE_____

NSN_____

CONTRACT NO_____

1. Break or cut off buttons at both ends.

2. Install immediately with direction of flow toward outlet.

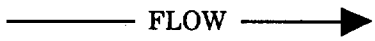


Figure 2.

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3.9 Reclaimed materials. The assemblies shall contain reclaimed materials to the maximum extent possible without jeopardizing the material quality or performance of the equipment. The reclaimed material shall be reprocessed, remanufactured, or recycled in a manner which restores them to the same chemical composition and physical properties as the material originally selected for use on the bit assemblies. Reclaimed materials shall be inclusive of all alloying elements applicable that have been collected from discarded solid, liquid, or gaseous waste from garbage, refuse, sludge, and from other collections of materials.

3.10 Workmanship. The cartridge shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, sharp edges, and accuracy of dimensions.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Certification of materials. All materials used in the construction of the cartridge shall be certified to meet the environmental requirements of 3.4. Filter media shall be certified to meet the filteration requirements of 3.6.2.

4.3 Classification of tests. The inspection and testing of the cartridge shall be classified as follows:

a. Preproduction tests (see 4.5).

b. Acceptance tests (see 4.6).

4.4 Test conditions.

4.4.1 Instrumentation.

4.4.1.1 Pressure measurements. Pressures shall be recorded in pounds per square inch gage (psig).

4.4.1.2 Temperature measurements. Temperatures shall be measured by appropriately located thermometers or thermocouples used with calibrated potentiometers. Temperatures shall be recorded in degrees Fahrenheit.

4.4.1.3 Flow measurements. Air flow measurements shall be obtained by passing the air through a properly calibrated meter designed for air. Air flow rates shall be recorded as standard cubic feet per minute (scfm). Total quantities of air passed shall be recorded as standard cubic feet (scf).

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4.4.1.4 Tolerances. The tolerances shall be as follows:

a. Pressure gages, air flow rates, and total flow quantities, to within 2 percent of the numerical figure recorded.

b. Temperatures shall be accurate to within 2° F.

c. Dewpoint shall be accurate to within 3° F.

4.4.2 Preparation for test. The cartridge shall be prepared for testing as required by the specific test instructions.

4.5 Preproduction testing.

4.5.1 Test samples. The preproduction test samples shall consist of four complete cartridge assemblies.

4.5.2 Preproduction tests. The preproduction tests shall consist of all the tests and examinations of 4.7 and shall be performed in the sequence as presented.

4.6 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests.

b. Sampling plans and tests.

4.6.1 Individual tests. Each cartridge shall be subjected to the following tests as described under 4.7:

a. Examination of product (see 4.7.1).

b. Two-minute leakage (see 4.7.3).

4.6.2 Sampling plans and tests.

4.6.2.1 Lot. A lot shall consist of cartridges manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

4.6.2.2 Sampling plan. Sampling plan test samples shall be selected in accordance with inspection level S-3 of MIL-STD-105 and shall be subjected to the following tests:

a. Dimensional examination (see 4.7.2).

b. Thirty minute leakage (see 4.7.4).

c. Quantity of desiccant (see 4.7.5).

4.7 Test methods.

4.7.1 Examination of product. The cartridge shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.7.2 Dimensions. The cartridge shall be checked to determine compliance with the dimensional requirements specified in Figure 1.

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4.7.3 Two-minute leakage. The cartridge shall be temperature stabilized throughout at 80° F or below immediately prior to this test. The cartridge shall be immersed in water maintained at a minimum temperature of 150° F for a minimum of 2 minutes to check for leakage. A definite continuous forming and breaking away of air bubbles from definite points on the surface of the cartridge shall be an indication of leakage and cause for rejection.

4.7.4 Thirty-minute leakage. The test of 4.7.3 shall be repeated except that the length of immersion shall be 30 minutes.

4.7.5 Quantity of desiccant. The cartridge shall be hand-held in a vertical plane and vigorously shaken up and down. Audible shifting or movement of the desiccant bed shall constitute failure of the test. When a cartridge fails to pass this test, four additional cartridges shall be tested. Failure of any of the additional four shall be cause for rejection of the lot. NOTE: Slight movement or rattling of the desiccant shall not be cause for rejection.

4.7.6 Shock test. The cartridge shall be subjected to the shock test as described in Procedure I, method 516, of MIL-STD-810.

4.7.7 Vibration test. The cartridge shall be subjected to a vibration test in accordance with Procedure IA of MIL-STD-810, with final test in vertical plane with outlet "down."

4.7.8 Performance test.

4.7.8.1 Apparatus. Apparatus for this test shall consist of a pressure vessel functionally designed to accept the cartridge and a 2 cfm source of 3,000 psig oil-free air, a saturator, a dew point meter, and air flow measuring apparatus. The dew point meter shall be accurate to within ±5° F and verification of the accuracy shall be furnished.

4.7.8.2 Conditions. The test shall consist of a continuous run until the dew point of the air leaving the cartridge reaches +40° F at 300 psig (approximately -20° F at 0 psig) or until a minimum of 75 standard (70° F and 760 mm Hg) cubic feet of air has been through the cartridge. The cartridge shall be weighed to an accuracy of ±0.2 grams immediately before and after each test run.

4.7.8.3 Test. The air shall be saturated with water vapor at a temperature of not less than 130° F and a pressure not greater than 300 psig. The saturated air shall then be passed through the sample cartridge at the above specified temperature and pressure and at the rate of 1.2 scfm. The dew point and quantity of flow shall be measured. The flow rate shall be constant to within ±1 percent.

4.7.8.3.1 Recorded data. The following readings shall be taken at intervals of not less than 2 minutes during the test run:

a. Saturator temperature and pressure

b. Temperature at inlet of cartridge

c. Dew point of air leaving cartridge

d. Rate of flow through cartridge

e. Total flow through cartridge (cumulative)

4.7.8.3.2 Capacity test. The total moisture content of the air leaving the cartridge shall not exceed that equivalent to a dew point of -65° F at 0 psig before a total flow of 75 standard (70° F and 760 mm Hg) cubic feet or the cartridge shall be considered to have failed the test.

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4.7.9 Flow test. Upon completion of the tests of 4.7.6, 4.7.7 and 4.7.8,the minimum flow of air through the cartridge mounted in a vertical position with the outlet end down and with a back pressure (input end of cartridge) of 15 psig shall be 2.0 scfm.

4.7.10 Visual examination after test. Upon completion of the tests of 4.7.6, 4.7.7 and 4.7.8, the cartridge shall be examined to determine that the filters have remained in place and that no by-pass of desiccant particles has occurred. NOTE: It may be necessary to dissect the cartridge and examine it internally to determine the above. Particular attention shall be paid to the output end of the cartridge downstream of the final filter to determine if desiccant particles have migrated around the filter media.

4.8 Inspection of the preservation, packaging, and packing. The inspection of the preservation, packaging, and packing shall be in accordance with the instructions of Section 5 herein.

5. PREPARATION FOR DELIVERY.

5.1 Preservation and packaging. Preservation and packaging shall be as specified in the contract or purchase order.

6. NOTES.

(This section contains information of a general or expanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cartridge covered by this specification is intended for use in dehydrating high-pressure air in airborne pneumatic systems.

6.2 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.3 Subject term (Key Word) listing.

Desiccant
Filtration
Moisture removal

Custodian:
Air Force - 99

Preparing activity:
Air Force - 84

Project No: 4440-F057

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-83996A (USAF)	Cartridge, Desiccant

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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